

popquiz power, conservation of energy , food Cal

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:04pm

Quiz Instructions

ONE ATTEMPT!!!



Question 1 17 pts

Each year 500 runners run up the stairs to the 86th floor of the Empire State Building in New York City. There are 1576 steps and each step is 0.28 m high. In 2003, Australian Paul Crake (20-29 age group) set the overall record by reaching the 86th floor in 9:33. His mass was 70.0 kg.

Say the time was 9 minutes . What is the energy he produced during this even? in J

hint: $PE = mgh$ and you have the height of each stair. take $g=10\text{m/s/s}$

1K is 1000 and 1M is one million

☐

about 309KJ

☐

about 200J

☐

about 31K

☐

about 1MJ



Question 2 17 pts

Building on the previous question. What is the energy in Cal (food calories)

1 Cal is 4184J.

round to nearest ten

☐

about 240Cal

☐

about 7000 Cal

☐

about 2000 Cal

☐

about 70 Cal



Question 3 17 pts

Building on the previous question

what is the power produced by the racer?

☐

about 200W

☐

about 1500W

☐

about 570W

☐

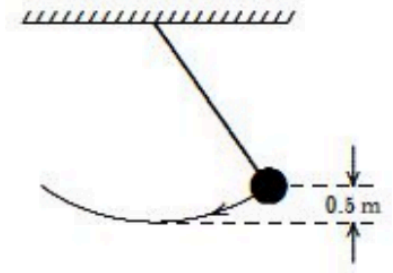
about 30,000 W



Question 4 17 pts

36. A simple pendulum consists of a 2.0-kg mass attached to a string. It is released from rest at X as shown. Its speed at the lowest point Y is about:

- ☐ (A) 0.90 m/s
- ☐ (B) 1.90 m/s
- ☐ (C) 3.1 m/s
- ☐ (D) 6.0 m/s
- ☐ (E) 36 m/s



use potential energy = kinetic energy

☐

A

☐

D

☐

B

☐

C

☐

E

⋮

Question 5 17 pts

In your body you have 7 Cal per gram.

So if you cut your Calories from 2,000 to 1,500 calories per day.

How many pounds do you lose per week ? (**take 450g = 1 pound about**)

hint: first find the grams. then the pounds

we supposed you have a good metabolism (no insulin resistance). There sre 7 days in a week

reference: Physics behind the headlines, Richard Muller

☐

about 0.5 pounds

☐

about 1 pound

☐

about 10pounds



about 5 pounds



Question 6 17 pts

in one half-hour of exercising , you lose 150 Cal about. (same Cal in a soda/orange juice).

If you exercise one hour everyday, how many grams you lose per week?

hint: use 7Cal in 1g.

(suppose a good metabolism. its an approximation . FYI one pound is about 500g)



about 500g



about 300g



about 1kg



about 700g



Question 7 5 pts

bonus

According to our discussion in class (and table at the end of the slides energy)

Note: we are talking about chocolate cookies. Energy in food can be found using a calorimeter.



You can not compare apples and oranges !



There is more energy in 1g of cookies than in 1 g of TNT (dynamite)



There is more energy in 1 g of TNT than in 1 g of cookies



I was on vacation that day



Question 8 4 pts

bonus

then why cookie eaters don't explode?

☐

There is no chemical reaction when cookies are digested.

☐

Its food, silly!

☐

Its not the same type of Joules

☐

The time to release of energy is larger for cookies

Not saved

Submit Quiz